

Group N: Final Report

Weibing Guo¹, Chloe Comeau², Yeteng Tang³, Cahit Erdem Anil⁴,
and Giovanni Picone⁵

¹University of Illinois Urbana-Champaign

²Vassar College

³University of Cambridge

⁴Bogaziçi University

⁵St. John's University

August 2025

Abstract

We classified links up to 11 crossings as ribbon or non-ribbon using four-genus, Jones polynomial, and symmetry criteria. Using existing data from the LinkInfo database, we eliminated as many non-ribbon links as possible before attempting to prove the remaining links were, in fact, ribbon. We concluded that since L10n32 and L10n36 both admit symmetric union presentations, they are ribbon. Moreover, they are the only two ribbon links with crossing number ≤ 11 .

1 Introduction

Ribbon links, a subclass of slice links, bound collections of disks immersed in the unit sphere and contain only ribbon singularities (Aceto et al.). The LinkInfo database contains a compendium of known information about links with up to 11 crossings; however, it currently contains no classifications of these links as ribbon or non-ribbon. We therefore set out to categorize all links in the database as ribbon or non-ribbon. We were able to narrow down the thousands of links with 11 crossings or fewer to identify the two ribbon links.

2 Methodology

We define slice links as links that bound collections of "disjoint, smoothly embedded disks" in four-space (Aceto et al.). Ribbon links are a subset of slice links. The four-genus of a link is 0 if and only if the link is slice (Powell). Thus, ribbon necessitates that a link's four-genus be 0. Similarly, the linking number between any two components of a link must be 0 for the link to be slice ((Aceto

et al.). This implies that all entries of the linking matrix must be 0. Therefore, if the linking matrix has any nonzero entries, the link cannot be slice, and hence cannot be ribbon.

Given the above requirements for ribbon links, we can eliminate all links but those with 0 linking matrices and smooth four-genus equal to 0. Using the advanced search feature on LinkInfo, we narrowed down the options.

From the thousands of links with crossing number less than or equal to 11, we narrowed our search to the following possibilities:

2-component links:

- L10n32
- L10n36
- L10n57
- L10n59

3-component links:

- L9n27
- L11n396
- L11n406
- L11n436

4-component links:

- L10n107

Unfortunately, this initial sweep failed to capture several invariants. Setting the smooth genus to 0 failed to capture the invariants L10n56, L11n244, L11n426, and L11n433. L10n56 and L11n244 both have four-genus $[0, 1]$, meaning they have genus of either 0 or 1, but no one has proven it to be definitively one or the other. L11n246 has four-genus $[0, 1]$ or 0, depending on the link's orientation. L11n433 has four-genus $[1, 2]$ or $[0, 1]$, also depending on orientation. All four links may have four-genus 0, so they are potentially ribbon links. However, since they have multiple possible 4-D genera, they were not captured in the initial search on LinkInfo. Adjusting our list to include the four additional invariants left us with a total of 13 potentially ribbon links.

For every n -component ribbon link L , the Jones polynomial $V(L)$ is divisible by the polynomial $V(\bigcirc^n)$ of the unlink (Eisermann). Thus, we can further reduce our list of potential ribbon links to the n -component links with Jones determinants divisible by $(q + q^{-1})^{n-1}$. Using SageMath and SnapPy, we found

the Jones polynomial for each link and assessed divisibility, generating the following results:

2-component links:

- L10n32: $(q + q^{-1})|V_L(q)$
- L10n36: $(q + q^{-1})|V_L(q)$
- L10n56: $(q + q^{-1})|V_L(q)$
- L10n57: $(q + q^{-1})|V_L(q)$
- L10n59: $(q + q^{-1})|V_L(q)$

3-component links:

- L9n27: $(q + q^{-1})^2 \nmid V_L(q)$
- L11n244: $(q + q^{-1})|V_L(q)$
- L11n396: $(q + q^{-1})^2 \nmid V_L(q)$
- L11n406: $(q + q^{-1})^2 \nmid V_L(q)$
- L11n426: $(q + q^{-1})^2 \nmid V_L(q)$
- L11n433: $(q + q^{-1})^2 \nmid V_L(q)$
- L11n436: $(q + q^{-1})^2 \nmid V_L(q)$

4-component links:

- L10n107: $(q + q^{-1})^3 \nmid V_L(q)$

After eliminating the links whose Alexander polynomials were not divisible by $(q + q^{-1})^{n-1}$, we were left with only the following potential ribbon links:

- L10n32: $(q + q^{-1})|V_L(q)$
- L10n36: $(q + q^{-1})|V_L(q)$
- L10n56: $(q + q^{-1})|V_L(q)$
- L10n57: $(q + q^{-1})|V_L(q)$
- L10n59: $(q + q^{-1})|V_L(q)$
- L11n244: $(q + q^{-1})|V_L(q)$

We then eliminated more invariants based on the modular equivalence of their determinants. Every n -component ribbon link $L = K_1 \cup \cdots \cup K_n$ satisfies $\det V(L) \equiv \det(K_1) \cdots \det(K_n) \pmod{32}$ (Eisermann).

The determinant of each n -component link is

$$\det(V(L)) = \frac{V(L)}{V(\bigcirc^n)_{q \rightarrow i}}$$

where $\bigcirc$ is the unknot, and $\bigcirc^n$ is the unlink with n unknots as components (Eisermann).

These are the determinants of the remaining potential ribbon links:

- L10n32: $\det V(L) = -23$
- L10n36: $\det V(L) = -23$
- L10n56: $\det V(L) = 17$
- L10n57: $\det V(L) = -15$
- L10n59: $\det V(L) = -23$
- L11n244: $\det V(L) = 33$

We used the modular equivalence theorem to examine each invariant.

L10n32

$$\det V(L) = -23$$

$$\det(K_1) = 9$$

$$\det(K_2) = 1$$

$$-23 \equiv 9 \cdot 1 = \det(K_1) \cdot \det(K_2)$$

L10n32 passes the test and is still potentially ribbon.

L10n36

$$\det V(L) = -23$$

$$\det(K_1) = 9$$

$$\det(K_2) = 1$$

$$-23 \equiv 9 \cdot 1 = \det(K_1) \cdot \det(K_2) \pmod{32}$$

L10n36 passes the test, so it is still possibly a ribbon link.

L10n56

$$\det V(L) = 17$$

$$\det(K_1) = 1$$

$$\det(K_2) = 1$$

$$17 \not\equiv 1 \cdot 1 \pmod{32}$$

L10n56 fails the test, so it cannot be ribbon.

L10n57

$$\det V(L) = -15$$

$$\det(K_1) = 1$$

$$\det(K_2) = 1$$

$$-15 \not\equiv 1 \cdot 1 \pmod{32}$$

L10n57 fails the test, so it cannot be ribbon.

L10n59

$$\det V(L) = -23$$

$$\det(K_1) = -3$$

$$\det(K_2) = -3$$

$$-23 \equiv 9 \cdot 1 \pmod{32}$$

L10n59 passes the test, so it remains a potential ribbon link.

L11n244

$$\det V(L) = 33$$

$$\det(K_1) = 1$$

$$\det(K_2) = -1$$

$$33 \not\equiv 1 \cdot (-1) \pmod{32}$$

L11n244 fails the test, so it cannot be ribbon.

Thus, we are left with only these three possibilities for ribbon links:

- L10n32
- L10n36
- L10n59

L10n59 is a two-component Conway link. Specifically, it is the pretzel link $P(2, -3, -2, 3)$. This link has a ribbon mutant but is not itself ribbon. Aceto et al. proved that two-component pretzel links of the form $P(p, q, -p, -q)$ are not slice (Aceto et al.). Since every ribbon knot is a slice knot (Teichner), L10n59 cannot be a ribbon link.

L10n32 and L10n36 both admit symmetric union presentations (see Figures 1 and 2, respectively). If a link admits a symmetric union presentation, it is a ribbon link (Lamm). Therefore, both are ribbon links.

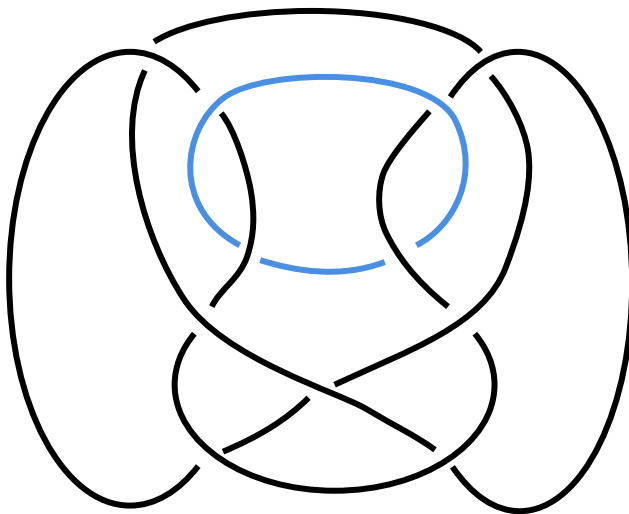


Figure 1: Symmetric Union Presentation of L10n32

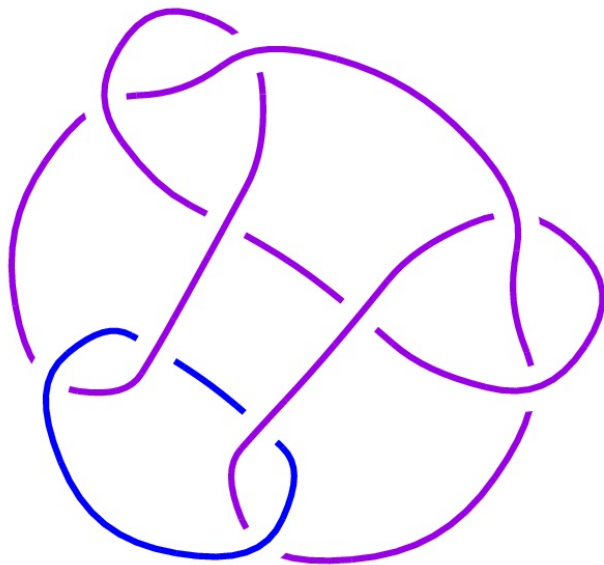


Figure 2: Symmetric Union Presentation of L10n36 (Livingston and Moore)

3 Results

Link Name	Ribbon (Yes/No)
L10n32	Yes
L10n36	Yes
L10n56	No
L10n57	No
L10n59	No
L9n27	No
L11n244	No
L11n396	No
L11n406	No
L11n426	No
L11n433	No
L11n436	No
L10n436	No

All other links with crossings less than or equal to 11 are not ribbon.

4 Conclusion

We developed and executed a categorization of all links with crossing number at most 11, combining several necessary conditions for ribbon links: smooth four-genus equals 0, all linking matrix entries are 0, link $L = K_1 \cup \cdots \cup K_n$ satisfies $\det V(L) \equiv \det(K_1) \cdots \det(K_n) \pmod{32}$, the Jones polynomial is divisible by $(q + q^{-1})^{m-1}$, and any two-component, slice pretzel link cannot take the form $P(p, q, -p, -q)$.

This multi-stage filtering eliminated candidates until only two links remained—L10n32 and L10n36. These two links were then proven to be ribbon since they admit symmetric union presentations. All other links fail at least one of the filtering criteria and are therefore not ribbon.

Works Cited

- Aceto, Paolo, et al. “Pretzel links, mutation, and the slice-ribbon conjecture”. *arXiv preprint arXiv:1805.02885*, 2018.
- Eisermann, Michael. “The Jones polynomial of ribbon links”. *Geometry & Topology*, vol. 13, no. 2, 2009, pp. 623–60.
- Lamm, Christoph. “Symmetric unions and ribbon knots”. 2000.
- Livingston, Charles, and Allison H. Moore. LinkInfo: Table of Link Invariants. Current Year.
- Powell, Mark. “The four-genus of a link, Levine–Tristram signatures and satellites”. *Journal of Knot Theory and Its Ramifications*, vol. 26, no. 02, 2017, p. 1740008.
- Teichner, Peter. “Slice knots: Knot theory in the 4th dimension”. *Lecture notes by Julia Collins and Mark Powell. Electronic version available from Andrew Ranicki’s homepage. <https://webhomes.maths.ed.ac.uk/~v1ranick/papers/sliceknots2.pdf>*, 2010.